2023
VCE
Specialist Mathematics
Year 12
Trial Examination 2

Detailed Answers

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SECTION A

ANSWERS

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SECTION A

Question 1 **Answer C**

If a function is continuous at a point, then it is differentiable at that point.
p: a function is continuous at a point
q: it is differentiable at that point.
$p \to q$ the contrapositive is $\neg q \to \neg p$
The contrapositive is
If a function is not differentiable at a point, then it is not continuous at that point.

Question 2 **Answer E**

$\int x^4 \cos(2x)\,dx$ using integration by parts $\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$

Let $u = x^4$, $\frac{dv}{dx} = \cos(2x)$

$\frac{du}{dx} = 4x^3$, $v = \int \cos(2x)\,dx = \frac{1}{2}\sin(2x)$

$\int x^4 \cos(2x)\,dx = \frac{x^4}{2}\sin(2x) - 2\int x^3 \sin(2x)\,dx$ Alan is correct

Alternatively let $u = \cos(2x)$, $\frac{dv}{dx} = x^4$

$\frac{du}{dx} = -2\sin(2x)$, $v = \int x^4 dx = \frac{x^5}{5}$

$\int x^4 \cos(2x)\,dx = \frac{x^5}{5}\cos(2x) + \frac{2}{5}\int x^5 \sin(2x)\,dx$ Ben is correct

From CAS, Colin is correct $\int x^4 \cos(2x)\,dx = \left(\frac{x^4}{2} - \frac{3x^2}{2} + \frac{3}{4}\right)\sin(2x) + \left(x^3 - \frac{3x}{2}\right)\cos(2x) + c$

$$\int\left(x^4 \cdot \cos(2\cdot x)\right)dx \qquad \frac{x\cdot\left(2\cdot x^2-3\right)\cdot\cos(2\cdot x)}{2} + \left(\frac{x^4}{2} - \frac{3\cdot x^2}{2} + \frac{3}{4}\right)\cdot\sin(2\cdot x)$$

Question 3 **Answer E**

The proposition statement $p(n)$: If n is even then $n^3 + 3$ is odd,
Can be shown to be true using a direct proof, an indirect proof using the contrapositive or a proof by contradiction, we cannot use mathematical induction to prove this proposition.

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Question 4 **Answer D**

$\underset{\sim}{a} = \underset{\sim}{i} + 2\underset{\sim}{j} - 2\underset{\sim}{k}$, $\underset{\sim}{b} = 2\underset{\sim}{i} - \underset{\sim}{j}$, $\underset{\sim}{c} = -\underset{\sim}{i} + 3\underset{\sim}{j} - 2\underset{\sim}{k}$

$\underset{\sim}{a} . \underset{\sim}{b} = 2 - 2 = 0$,

so the vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are perpendicular

A. is true

$\underset{\sim}{a} \times \underset{\sim}{c} = \underset{\sim}{b} \times \underset{\sim}{c} = 2\underset{\sim}{i} + 4\underset{\sim}{j} + 5\underset{\sim}{k}$ **B.** is true

$\underset{\sim}{c} = \underset{\sim}{a} - \underset{\sim}{b}$ so the vectors $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{c}$ are linearly dependent. **C.** is true

$|\underset{\sim}{a}| = 3$, $|\underset{\sim}{b}| = \sqrt{5}$, $|\underset{\sim}{c}| = \sqrt{14}$

$|\underset{\sim}{b}| < |\underset{\sim}{a}|$ so $|\underset{\sim}{b}| > |\underset{\sim}{a}|$ **D.** is false

$|\underset{\sim}{c}| > |\underset{\sim}{b}|$ **E.** is true

$[1\ \ 2\ \ -2] \to a$	$[1\ \ 2\ \ -2]$
$b:=[2\ \ -1\ \ 0]$	$[2\ \ -1\ \ 0]$
Define $c=[-1\ \ 3\ \ -2]$	*Done*
$\text{dotP}(a,b)$	0
$\text{crossP}(a,c)$	$[2\ \ 4\ \ 5]$
$\text{crossP}(b,c)$	$[2\ \ 4\ \ 5]$
$\text{solve}(c=m\cdot a+n\cdot b,\{m,n\})$	$m=1$ and $n=-1$
$\text{norm}(a)$	3
$\text{norm}(b)$	$\sqrt{5}$
$\text{norm}(c)$	$\sqrt{14}$

Question 5 **Answer A**

Assume $\left(r\,\text{cis}(\theta)\right)^k = r^k\,\text{cis}(k\theta)$ and show that $\left(r\,\text{cis}(\theta)\right)^{k+1} = r^{k+1}\,\text{cis}\left((k+1)\theta\right)$

Question 6 **Answer C**

$u = -3 - 3i = 3\sqrt{2}\,\text{cis}\left(\dfrac{-3\pi}{4}\right)$

$u^n = \left(3\sqrt{2}\right)^n \text{cis}\left(\dfrac{-3n\pi}{4}\right)$

$u^n = 18^{\frac{n}{2}}\left(\cos\left(\dfrac{-3n\pi}{4}\right) + i\sin\left(\dfrac{-3n\pi}{4}\right)\right)$

$u^n = \sqrt{18^n}\left(\cos\left(\dfrac{3n\pi}{4}\right) - i\sin\left(\dfrac{3n\pi}{4}\right)\right)$

If $n = 4k$, $n = 4, 8, 12, ..$ and $k \in Z^+$

then $\sin(3\pi k) = 0$

so that u^n is purely real. **A.** is true

If $n = 4k - 2, n = 2, 6, 10, ...$ and $k \in Z^+$

then $\cos\left(\dfrac{3\pi}{2}(k-1)\right) = 0$ so that u^n is purely imaginary. **B.** is true

$u:=-3-3\cdot i$	$-3-3\cdot i$
u^2	$18\cdot i$
u^3	$54-54\cdot i$
u^4	-324
u^5	$972+972\cdot i$
u^6	$-5832\cdot i$
u^7	$-17496+17496\cdot i$
u^8	104976
u^9	$-314928-314928\cdot i$
u^{10}	$1889568\cdot i$
u^{11}	$5668704-5668704\cdot i$
u^{12}	-34012224

$|u^n| = \sqrt{18^n}$ **D.** is true and $\arg(u^n) = -\dfrac{3\pi n}{4}$, **E.** is true, when $n = 2k+1$, $n = 3, 5, 7, 9, ..$ **C.** is false

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Question 7 **Answer B**

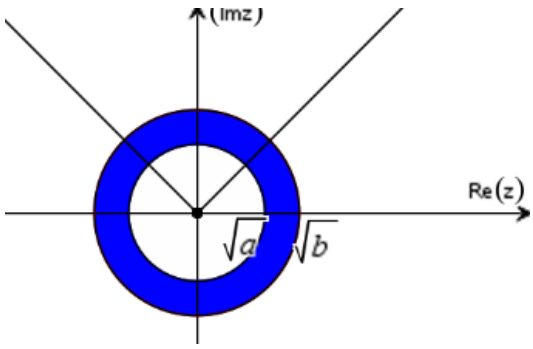
$\{z : a < z\bar{z} \le b, z \in C\} \bigcap \{z : \frac{\pi}{4} \le \text{Arg}(z) \le \frac{3\pi}{4}, z \in C\}$

$z = x + yi, \quad \bar{z} = x - yi,$

$z.\bar{z} = (x+yi).(x-yi) = x^2 - y^2 i^2$

$z.\bar{z} = x^2 + y^2 = r^2$

concentric circles of radii $\sqrt{a}$ and $\sqrt{b}$,
but it is only one-quarter of the area of the region between the circles, so that $A = \frac{\pi}{4}(b-a)$



Question 8 **Answer C**

The distance of a plane $ax + by + cz = d$ from the origin is $\frac{d}{\sqrt{a^2+b^2+c^2}}$

plane (1) $\underset{\sim}{r} = x\underset{\sim}{i} + y\underset{\sim}{j} + z\underset{\sim}{k}$, $\underset{\sim}{n} = 4\underset{\sim}{i} + 2\underset{\sim}{j} - 4\underset{\sim}{k}$, $\underset{\sim}{r}.\underset{\sim}{n} = 4$ gives $4x + 2y - 4z = 4$ or

$2x + y - 2z = 2$, the distance of this plane from the origin is $\frac{2}{\sqrt{2^2 + 1^2 + (-2)^2}} = \frac{2}{3}$

(2) $-2x - y + 2z = 1$ distance of this plane from the origin $\frac{1}{\sqrt{(-2)^2 + (-1)^2 + 2^2}} = \frac{1}{3}$

but their normals are pointing in the opposite directions, so the planes lie on opposite sides of the origin, so the distance between them is $\frac{2}{3} + \frac{1}{3} = 1$

Question 9 **Answer D**

$x = t - \sin(t)$ $\quad y = 1 - \cos(t) = 2\sin^2\left(\frac{t}{2}\right)$ $\quad t \in [0, 2\pi], \quad t_1 = 0, \; t_2 = 2\pi$

$\frac{dx}{dt} = 1 - \cos(t) = 2\sin^2\left(\frac{t}{2}\right)$, $\frac{dy}{dt} = -\sin(t) = -2\sin\left(\frac{t}{2}\right)\cos\left(\frac{t}{2}\right)$

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 4\sin^4\left(\frac{t}{2}\right) + 4\sin^2\left(\frac{t}{2}\right)\cos^2\left(\frac{t}{2}\right) = 4\sin^2\left(\frac{t}{2}\right)\left(\sin^2\left(\frac{t}{2}\right) + \cos^2\left(\frac{t}{2}\right)\right) = 4\sin^2\left(\frac{t}{2}\right)$$

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \left|2\sin\left(\frac{t}{2}\right)\right| = 2\sin\left(\frac{t}{2}\right) \quad \text{since } \frac{t}{2} \in [0, \pi]$$

$$S = 2\pi \int_{t_1}^{t_2} y(t)\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$S = 8\pi \int_0^{2\pi} \sin^3\left(\frac{t}{2}\right) dt$$

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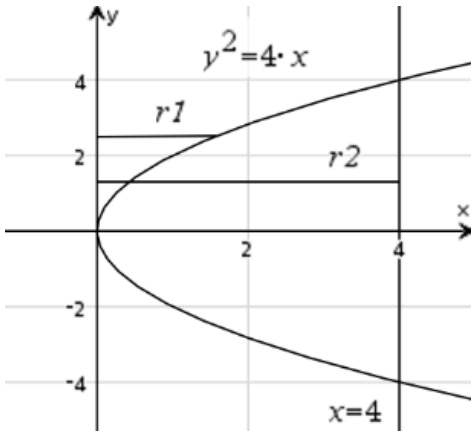
Question 10 **Answer E**



$$V=\pi\int_a^b\left(r_2^2-r_1^2\right)dy$$

$$r_2=4,\quad r_1=x=\frac{y^2}{4}$$

$$V=\pi\int_{-4}^{4}\left(16-\frac{y^4}{16}\right)dy$$

$$V=2\pi\int_0^4\left(16-\frac{y^4}{16}\right)dy$$

Question 11 **Answer D**

The distance of the point $A(2,1,3)$ to the line

$\frac{x-3}{2}=\frac{y+2}{2}=1-z$ or $\frac{x-3}{2}=\frac{y+2}{2}=\frac{z-1}{-1}$

Now a point on the line is $P_0(3,-2,1)$ and the direction of the line $\underset{\sim}{v}=2\underset{\sim}{i}+2\underset{\sim}{j}-\underset{\sim}{k}$, $|\underset{\sim}{v}|=3$

$$\overrightarrow{AP_0}=\overrightarrow{OP_0}-\overrightarrow{OA}=\underset{\sim}{i}-3\underset{\sim}{j}-2\underset{\sim}{k}$$

$$D=\left|\overrightarrow{AP_0}\times\hat{\underset{\sim}{v}}\right|=\frac{\sqrt{122}}{3}$$

$ap0:=[1\ \ -3\ \ -2]$	$[1\ \ -3\ \ -2]$
$v:=[2\ \ 2\ \ -1]$	$[2\ \ 2\ \ -1]$
$\text{norm}(\text{crossP}(\text{unitV}(v),ap0))$	$\frac{\sqrt{122}}{3}$

Question 12 **Answer B**

The distance of the point to the plane.

Let $A(2,1,3)$, $x_0=2,\ y_0=1,\ z_0=3$ to the plane $2x-y+2z=6$,

the normal to the plane is $\underset{\sim}{n}=2\underset{\sim}{i}-\underset{\sim}{j}+2\underset{\sim}{k}$, $a=2,\ b=-1,\ c=2,\ d=6$

the distance is $D=\left|\frac{ax_0+by_0+cz_0-d}{\sqrt{a^2+b^2+c^2}}\right|=\left|\frac{4-1+6-6}{\sqrt{4+1+4}}\right|=1$

Question 13 **Answer A**

The distance of the line to the plane is the same as the distance of a point on the line to the plane.

Let $P_0(3,-2,1)$ be a point on the line, $x_0=3,\ y_0=-2,\ z_0=1$,

the normal to the plane is $\underset{\sim}{n}=2\underset{\sim}{i}-\underset{\sim}{j}+2\underset{\sim}{k}$, $a=2,\ b=-1,\ c=2,\ d=6$

$$D=\left|\frac{ax_0+by_0+cz_0-d}{\sqrt{a^2+b^2+c^2}}\right|=\left|\frac{6+2+2-6}{\sqrt{4+1+4}}\right|=\frac{4}{3}$$

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Question 14 **Answer E**

$$\frac{dy}{dx} = 2x\tan(y) \qquad y(-1) = \frac{\pi}{3}$$

$$\left.\frac{dy}{dx}\right|_{\left(-1,\frac{\pi}{3}\right)} = -2\tan\left(\frac{\pi}{3}\right) = -2\sqrt{3}$$

using implicit differentiation

$$\frac{d^2y}{dx^2} = 2\tan(y) + 2x\sec^2(y)\frac{dy}{dx}$$

$$\left.\frac{d^2y}{dx^2}\right|_{\left(-1,\frac{\pi}{3}\right)} = 2\tan\left(\frac{\pi}{3}\right) - 2\sec^2\left(\frac{\pi}{3}\right)\times -2\sqrt{3}$$

$$= 2\sqrt{3} + 16\sqrt{3} = 18\sqrt{3}$$

$$\text{deSolve}\left(y'=2\cdot x\cdot\tan(y) \text{ and } y(-1)=\frac{\pi}{3}, x, y\right) \qquad \frac{\ln\left(\frac{4\cdot(\sin(y))^2}{3}\right)}{2}=x^2-1$$

$$\text{solve}\left(\frac{\ln\left(\frac{4\cdot(\sin(y))^2}{3}\right)}{2}=x^2-1, y\right)$$

$$y=-\sin^{-1}\left(\frac{\sqrt{3}\cdot e^{x^2-1}}{2}\right)+2\cdot n8\cdot\pi+\pi \text{ and } e^{x^2}\le\frac{2\cdot e\cdot\sqrt{3}}{3} \text{ or } y=2\cdot n9\cdot\pi-\sin^{-1}\left(\frac{\sqrt{3}}{}\right.$$

$$\frac{d^2}{dx^2}\left(\sin^{-1}\left(\frac{\sqrt{3}\cdot e^{x^2-1}}{2}\right)\right)\Big|_{x=-1} \qquad 18\cdot\sqrt{3}$$

Question 15 **Answer A**

$$\ddot{x} = g - kv, \qquad \ddot{x} = 0, \qquad v_T = \frac{g}{k}$$

$$\ddot{x} = \frac{dv}{dt} = g - kv, \qquad \frac{dt}{dv} = \frac{1}{g-kv}$$

$$T = \int_{1.5v_T}^{1.1v_T} \frac{1}{g-kv}\,dv = \frac{1}{k}\log_e(5)$$

$$\ddot{x} = v\frac{dv}{dx} = g - kv, \qquad \frac{dx}{dv} = \frac{v}{g-kv}$$

$$D = \int_{1.5v_T}^{1.1v_T} \frac{v}{g-kv}\,dv$$

$$D = \frac{g}{5k^2}\left(5\log_e(5)+2\right)$$

$$\int_{1.5\cdot tv}^{1.1\cdot tv} \frac{1}{g-k\cdot v}\,dv\Big|_{tv=\frac{g}{k}} \qquad \frac{\ln(5)}{k}$$

$$\int_{1.5\cdot tv}^{1.1\cdot tv} \frac{v}{g-k\cdot v}\,dv\Big|_{tv=\frac{g}{k}} \qquad \frac{g\cdot(5\cdot\ln(5)+2)}{5\cdot k^2}$$

Question 16 **Answer A**

when $y = x$ $m = 1$, when $y = -x$ $m = -1$

when $y = 0$ $m = -\frac{1}{2}$, when $x = 0$ $m = -2$

when $y = \frac{x}{2}$ $m = 0$, when $y = 2x$ $m = 0$

these are only satisfied by $\frac{dy}{dx} = \frac{x-2y}{y-2x}$

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Question 17 **Answer D**

$$\frac{dy}{dx} = f(x,y) = 2xy^2,$$
$$x_0 = 1,\ y_0 = 1,\ h = 0.25,\ x_1 = 1.25$$
$$y_1 = y_0 + hf(x_0, y_0)$$
$$y_1 = 1 + 0.25\left(2\times1\times1^2\right) = 1.5$$
$$y_2 = y_1 + hf(x_1, y_1)$$
$$y_2 = 1.5 + 0.25\left(2\times1.25\times1.5^2\right) = 2.906$$

$$\text{euler}\left(2\cdot x\cdot y^2, x, y, \{1,2\}, 1, 0.25\right)$$
$$\begin{bmatrix} 1.0000 & 1.2500 & 1.5000 & 1.7500 & 2.0000 \\ 1.0000 & 1.5000 & 2.9063 & 9.2410 & 83.9620 \end{bmatrix}$$

Question 18 **Answer C**

$$95\% \quad z = 1.96, \quad s = 12, \quad n = ? \quad \left(\bar{x} - z\frac{s}{\sqrt{n}},\ \bar{x} + z\frac{s}{\sqrt{n}}\right) = (246.1, 253.9)$$

$$2z\frac{s}{\sqrt{n}} = 253.9 - 246.1 = 7.8, \qquad n = \left(\frac{2zs}{7.8}\right)^2 = \left(\frac{2\times1.96\times12}{7.8}\right)^2 = 36$$

Question 19 **Answer B**

$$C\% \quad z = ?, \quad s = 20, \quad n = 36, \quad \left(\bar{x} - z\frac{s}{\sqrt{n}},\ \bar{x} + z\frac{s}{\sqrt{n}}\right) = (494, 506)$$

$$2z\frac{s}{\sqrt{n}} = 506 - 494 = 12$$
$$z = \frac{12\sqrt{n}}{2s} = \frac{6\times6}{20} = 1.8$$
$$Z \overset{d}{=} N(0,1), \quad \Pr(-1.8 < Z < 1.8) = 0.93$$

normCdf(-1.8,1.8,0,1)	0.9281

Question 20 **Answer B**

$$C \overset{d}{=} N\left(125, 10^2\right) \quad P \overset{d}{=} N\left(185, 15^2\right)$$
$$D = C_1 + C_2 + C_3 - (P_1 + P_2) = 3C - 2P$$
$$E(D) = 3E(C) - 2E(P)$$
$$= 3\times125 - 2\times185 = 5$$
$$\text{Var}(D) = 3\text{Var}(C) + 2\text{Var}(P)$$
$$= 3\times10^2 + 2\times15^2 = 750$$
$$D \overset{d}{=} N(5, 750)$$
$$\Pr(-2 < D < 2) = 0.0573$$

$3\cdot125-2\cdot185$	5
$3\cdot10^2+2\cdot15^2$	750
$\text{normCdf}\left(-2,2,5,\sqrt{750}\right)$	0.0573

END OF SECTION A SUGGESTED ANSWERS

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SECTION B

Question 1

a. $f(x)=\dfrac{x^2-3}{x^2-4x+3}=\dfrac{\left(x-\sqrt{3}\right)\left(x+\sqrt{3}\right)}{(x-3)(x-1)}=1+\dfrac{2(2x-3)}{(x-3)(x-1)}$

the graph has $y=1$ as a horizontal asymptote
and $x=1$ and $x=3$ as vertical asymptotes. A1

b. $f'(x)=\dfrac{-4\left(x^2-3x+3\right)}{\left(x^2-4x+3\right)^2}$ since $x^2-3x+3\neq 0$

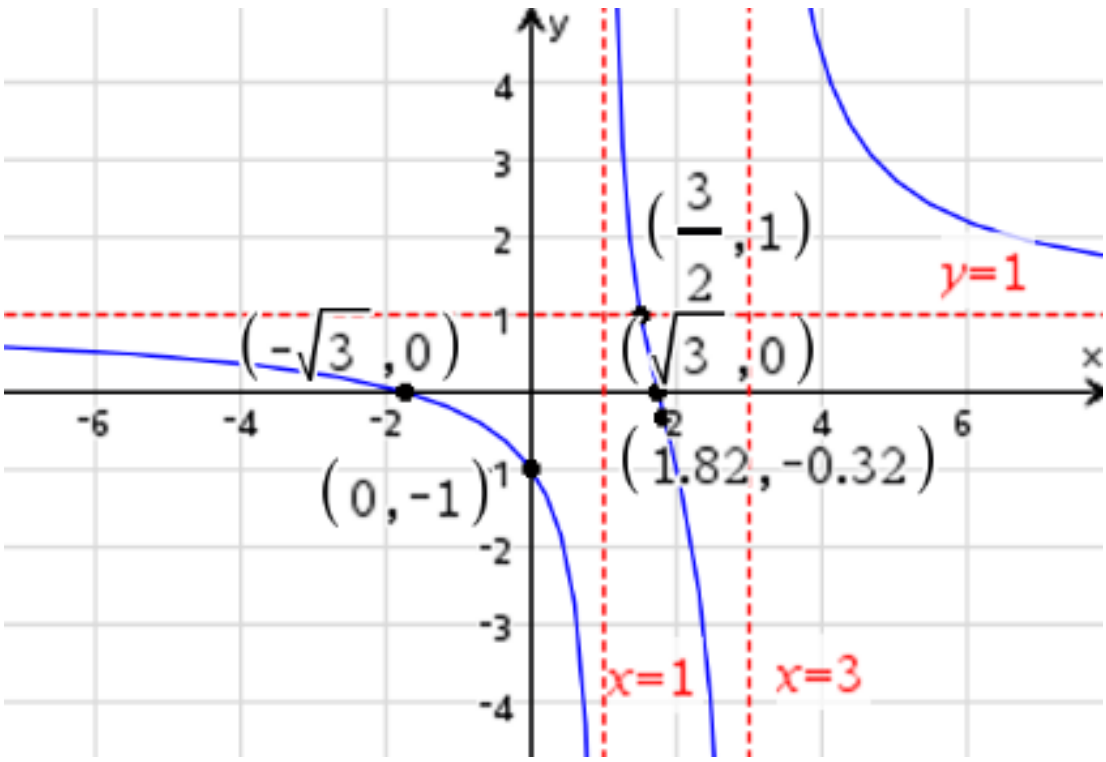
the discriminant $\Delta=9-4\times 3=-3<0$ A1
so that $f'(x)\neq 0$ therefore the graph has no turning points

c. the point of inflexion occurs when $f''(x)=\dfrac{4\left(2x^3-9x^2+18x-15\right)}{\left(x^2-4x+3\right)^3}=0$

solving $2x^3-9x^2+18x-15=0$ gives $x=1.819$, $f(1.819)=-0.319$,
one point of inflexion at $(1.82,-0.32)$

the graph crosses the horizontal asymptote when $y=1$, $x=\dfrac{3}{2}$, $\left(\dfrac{3}{2},1\right)$

axial intercepts $\left(\pm\sqrt{3},0\right)(0,-1)$ all features of the graph G2



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$$f(x)=\frac{x^2-c}{x^2-4x+c}=1+\frac{2(2x-c)}{x^2-4x+c}$$

d.i. When $c=0$, $f(x)=\dfrac{x^2}{x^2-4x}=\dfrac{x}{x-4}=1+\dfrac{4}{x-4}$ hyperbola

there is a point of discontinuity at $x=0$.

When $c=4$, $f(x)=\dfrac{x^2-4}{x^2-4x+4}=\dfrac{(x+2)(x-2)}{(x-2)^2}=\dfrac{x+2}{x-2}=1+\dfrac{4}{x-2}$ hyperbola A1

there is a point of discontinuity at $x=2$.
When the graph is a hyperbola, there are no turning points and no points of inflexion.
When $c=0$ there is one vertical asymptote at $x=4$ and when $c=4$, $x=2$ is a point of discontinuity, but there is still a vertical asymptote at $x=2$.

d.ii. The graph crosses the *x*-axis twice when $x=\pm\sqrt{c}$ when $c>0$, but $c\neq 4$.
The graph will cross the *x*-axis once when $c=4$.
The graph will not cross the *x*-axis when $c<0$

A1

The graph of $y=f(x)$	values of c
crosses the *x*-axis twice	$(0,4)\cup(4,\infty)$
crosses the *x*-axis once	$c=4$
does not cross the *x*-axis	$(-\infty,0)$

d.iii. The graph has vertical asymptotes when the denominator is zero.
So $x^2-4x+c=0$, $\Delta=16-4c$, $x=2\pm\sqrt{4-c}$
Two vertical asymptotes when $\Delta>0$, $c<4$ but not when $c=0$
One vertical asymptote when $\Delta=16-4c=0$, $c=4$ but also when $c=0$
No vertical asymptotes when $\Delta<0$, $c>4$ A1

The graph of $y=f(x)$	values of c
has two vertical asymptotes	$(-\infty,0)\cup(0,4)$
has one vertical asymptote	$c=0$ or $c=4$
has no vertical asymptotes	$(4,\infty)$

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d.iv. turning points occur when $f'(x)=\dfrac{-4\left(x^2-cx+c\right)}{\left(x^2-4x+c\right)^2}=0$

$x^2-cx+c=0,\quad \Delta=c^2-4c=c(c-4)$

For two turning points $\Delta>0,\ \ c>4$ or $c<0$

For no turning points $\Delta<0,\ \ 0<c<4$ but also no turning point when $c=0$ or $c=4$

A2

The graph of $y=f(x)$	values of c
has no turning points	$[0,4]$
has one maximum and one minimum turning point	$(-\infty,0)\cup(4,\infty)$

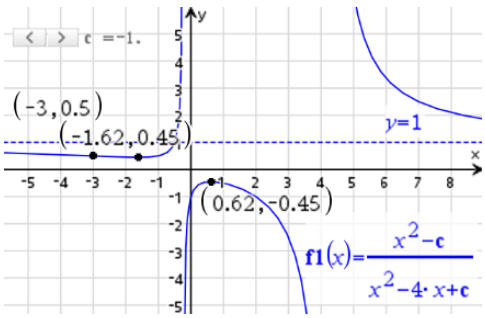
d.v. No inflexion points occur when $c=0$ or $c=4$

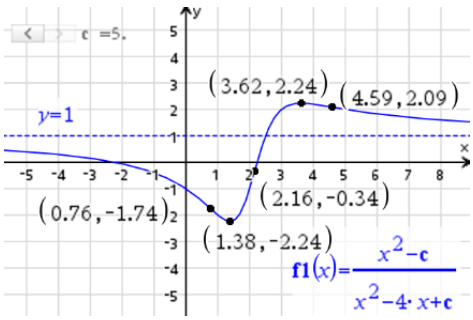
One inflexion point occurs when the graph has two vertical asymptotes $c\in(-\infty,0)\cup(0,4)$.

There is never two points of inflexion, and the graph has three points of inflexion when it has no vertical asymptotes, $c>4$

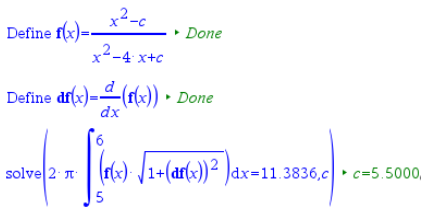
A2

The graph of $y=f(x)$	values of c
has no points of inflexion	$c=0$ or $c=4$
has one point of inflexion	$(-\infty,0)\cup(0,4)$
has two points of inflexion	no values of c
has three points of inflexion	$(4,\infty)$





e.



solving

$2\pi\int_5^6 f(x)\sqrt{1+\left(f'(x)\right)^2}\,dx=11.3836$ A1

$c=5.5$

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Question 2 $u = 2 - i$

a. $u^3 = 2 - 11i, \quad P(u^3) = 0 \Rightarrow (z - u^3) = (z - 2 + 11i)$ is a factor

$P(z) = (z - 2 + 11i)(z^2 + k) = z^3 + vz^2 + 5z + w = 0$

$\Rightarrow k = 5$ M1

$P(z) = (z - 2 + 11i)(z^2 + 5) = z^3 + (-2 + 11i)z^2 + 5z + 55i - 10 = 0$

$v = -u^3 = -2 + 11i$ A1

$w = -10 + 55i$

b. $Q(2 - i) = 0$ since a,b,c and d are real, by the conjugate root theorem $Q(2 + i) = 0$

$u.\bar{u} = 4 - i^2 = 5, \quad u + \bar{u} = 4$

therefore $z^2 - 4z + 5$ is a quadratic factor, now $u^2 = 3 - 4i$

$Q(3 - 4i) = 0$ and $Q(3 + 4i) = 0$

$u^2.\bar{u}^2 = 9 - 16i^2 = 25, \quad u^2 + \bar{u}^2 = 6$

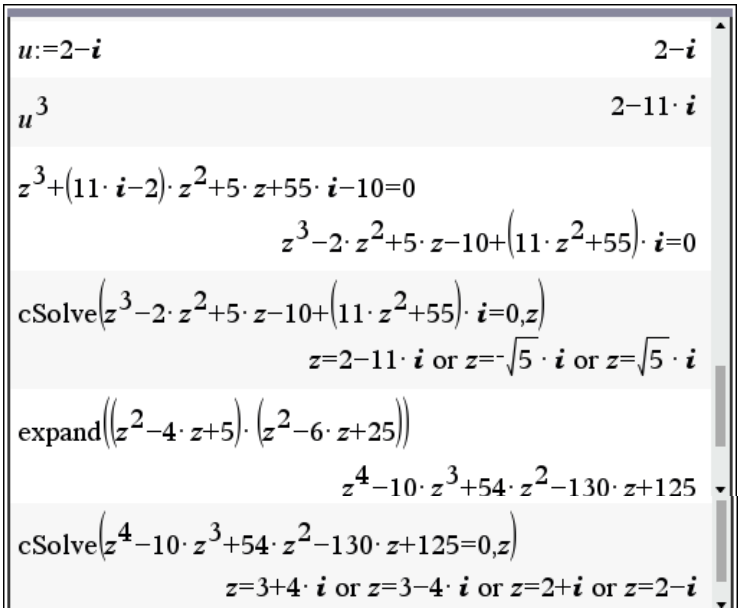
therefore $z^2 - 6z + 25$ is another quadratic factor M1

$Q(z) = z^4 + az^3 + bz^2 + cz + d$

$Q(z) = (z^2 - 4z + 5)(z^2 - 6z + 25)$ expanding

$Q(z) = z^4 - 10z^3 + 54z^2 - 130z + 125$

$a = -10, \quad b = 54, \quad c = -130, \quad d = 125$ A1



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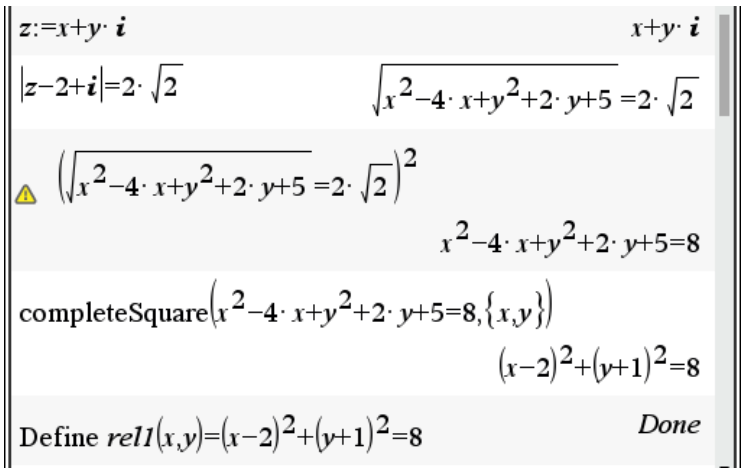
c. Let $z = x + yi, \quad |z - u| = 2\sqrt{2}$

$|x - 2 + (y+1)i| = 2\sqrt{2}$

$\sqrt{(x-2)^2 + (y+1)^2} = 2\sqrt{2}$

$(x-2)^2 + (y+1)^2 = 8$

A circle with centre at $(2,-1)$ and radius $2\sqrt{2}$. A1



d. $\text{Arg}(z-u) = \text{Arg}(x - 2 + (y+1)i) = \dfrac{3\pi}{4}$

$\tan^{-1}\left(\dfrac{y+1}{x-2}\right) = \dfrac{3\pi}{4}, \quad x < 2$

$\dfrac{y+1}{x-2} = \tan\left(\dfrac{3\pi}{4}\right) = -1$

$y + 1 = -x + 2$

$y = 1 - x, \quad x < 2$

A ray from the point $(2,-1)$ not including this point, making an angle of 135^0 with the positive real axis. A1

e. solving $(x-2)^2 + (y+1)^2 = 8$ and $y = 1 - x, \quad x < 2$, gives

$(x-2)^2 + (2-x)^2 = 8$

$2(x-2)^2 = 8, \quad (x-2)^2 = 4$

$x - 2 = \pm 2$

$x = 4, 0,$ but $x < 2$

so $x = 0, \ y = 1,$ only $(0,1)$ A1

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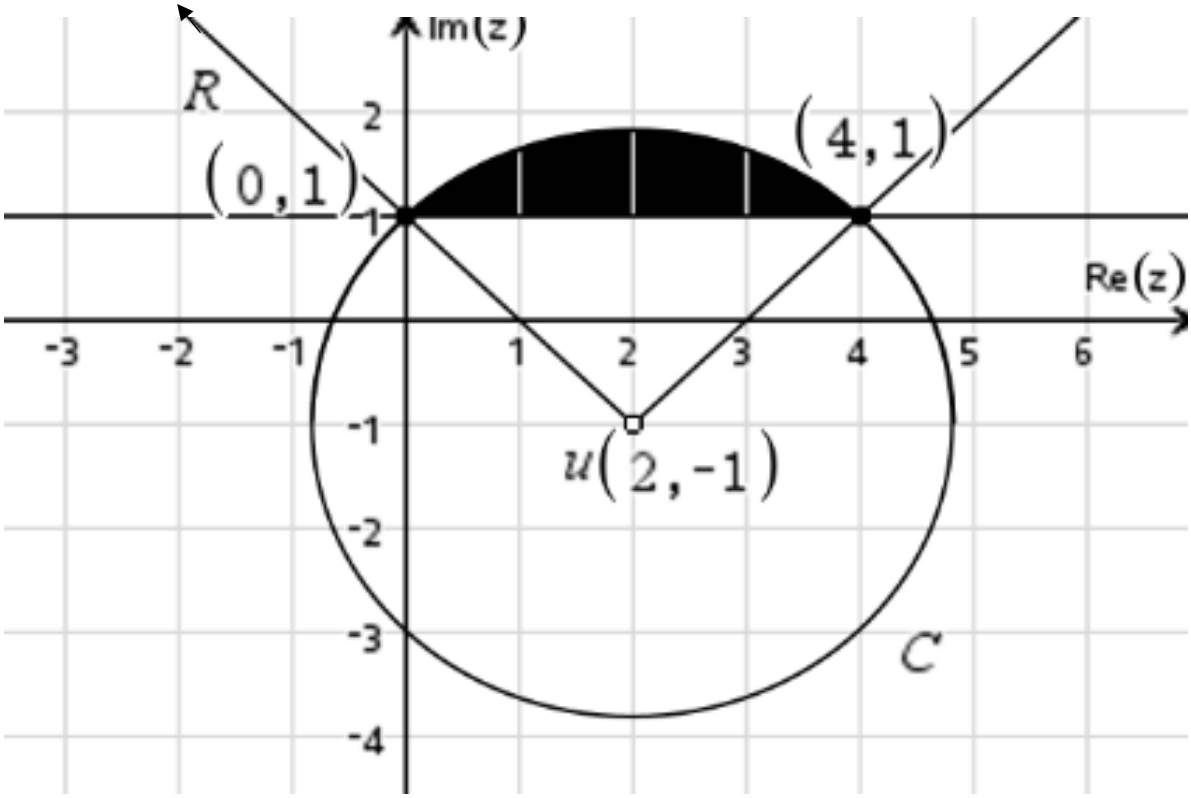
$\text{angle}(z-2+i)=\frac{3\cdot\pi}{4}$	$\frac{\pi\cdot\text{sign}(y+1)}{2}-\tan^{-1}\left(\frac{x-2}{y+1}\right)=\frac{3\cdot\pi}{4}$
$\text{solve}\left(\frac{\pi\cdot\text{sign}(y+1)}{2}-\tan^{-1}\left(\frac{x-2}{y+1}\right)=\frac{3\cdot\pi}{4},y\right)$	$y=1-x$ and $x<2$
$\text{solve}\left((x-2)^2+(y+1)^2=8 \text{ and } y=1-x,\{x,y\}\right)\lvert x<2$	$x=0$ and $y=1$
Define $f1(x)=1-x\lvert x<2$	*Done*

f.g. must have an open circle at u. G3



h. the angle is $\theta=\frac{\pi}{2}$ and the radius $r=2\sqrt{2}$

the area of the sector is $A=\frac{r^2}{2}(\theta-\sin(\theta))=4\left(\frac{\pi}{2}-1\right)=2\pi-4$ A1

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Question 3

a. $A(4,5,-2)$ $B(6,2,-3)$ and $C(a,b,0)$ where $a,b \in Z$.

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 2\underset{\sim}{i} - 3\underset{\sim}{j} - \underset{\sim}{k}$ $\quad \left|\overrightarrow{AB}\right| = \sqrt{4+9+1} = \sqrt{14}$

$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = (a-6)\underset{\sim}{i} + (b-2)\underset{\sim}{j} + 3\underset{\sim}{k}$ $\quad \left|\overrightarrow{AB}\right| = \sqrt{(a-6)^2 + (b-2)^2 + 9}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = (a-4)\underset{\sim}{i} + (b-5)\underset{\sim}{j} + 2\underset{\sim}{k}$ $\quad \left|\overrightarrow{AC}\right| = \sqrt{(a-4)^2 + (b-5)^2 + 4}$ M1

since the triangle is equilateral all side lengths are equal, so that $\left|\overrightarrow{AB}\right| = \left|\overrightarrow{BC}\right| = \left|\overrightarrow{AC}\right| = \sqrt{14}$,

hence squaring gives

(1) $(a-6)^2 + (b-2)^2 = 5$

(2) $(a-4)^2 + (b-5)^2 = 10$

verifying (1) and (2) are satisfied when $a = 7$, $b = 4$ A1

b. $\overrightarrow{AC} = 3\underset{\sim}{i} - \underset{\sim}{j} + 2\underset{\sim}{k}$

$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 2 & -3 & -1 \\ 3 & -1 & 2 \end{vmatrix} = -7\underset{\sim}{i} - 7\underset{\sim}{j} + 7\underset{\sim}{k}$ and $\left|\overrightarrow{AB} \times \overrightarrow{AC}\right| = 7\sqrt{3}$ A1

so a normal to the plane is $\underset{\sim}{n} = -\underset{\sim}{i} - \underset{\sim}{j} + \underset{\sim}{k}$, the equation of the plane through A, is

$-(x-4) - (y-5) + (z+2) = 0$

$x + y - z = 11$

The area of the triangle is $\frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right| = \frac{7\sqrt{3}}{2}$ A1

c. $A(4,5,-2)$ $B(6,2,-3)$ and $C(7,4,0)$

midpoints $D = \frac{1}{2}(A+B)$, $\quad E = \frac{1}{2}(B+C)$, $F = \frac{1}{2}(A+C)$

$D\left(5, \frac{7}{2}, -\frac{5}{2}\right)$, $\quad E\left(\frac{13}{2}, 3, -\frac{3}{2}\right)$, $\quad F\left(\frac{11}{2}, \frac{9}{2}, -1\right)$ A1

d. $\overrightarrow{AG} = \frac{2}{3}\overrightarrow{AE}$, $\overrightarrow{BG} = \frac{2}{3}\overrightarrow{BF}$ and $\overrightarrow{CG} = \frac{2}{3}\overrightarrow{CD}$

$G\left(\frac{17}{3}, \frac{11}{3}, -\frac{5}{3}\right)$, note that all of D, E, F and G all lie in the plane A1

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e. normal to the plane $\underset{\sim}{n} = -\underset{\sim}{i} - \underset{\sim}{j} + \underset{\sim}{k}$, so the line through $G\left(\frac{17}{3}, \frac{11}{3}, -\frac{5}{3}\right)$, is

$$\frac{x-\frac{17}{3}}{-1} = \frac{y-\frac{11}{3}}{-1} = \frac{z+\frac{5}{3}}{1} = t \quad \Rightarrow \quad x = \frac{17}{3} - t, \quad y = \frac{11}{3} - t, \quad z = -\frac{5}{3} + t$$ A1

f. let the coordinates of V be $V(x, y, z)$ since $d\left(\overrightarrow{AV}\right) = d\left(\overrightarrow{BV}\right) = d\left(\overrightarrow{CV}\right) = \sqrt{61}$

$eq1 := (x-4)^2 + (y-5)^2 + (z+2)^2 = 61$

$x^2 - 8\cdot x + y^2 - 10\cdot y + z^2 + 4\cdot z + 45 = 61$

$eq2 := (x-6)^2 + (y-2)^2 + (z+3)^2 = 61$

$x^2 - 12\cdot x + y^2 - 4\cdot y + z^2 + 6\cdot z + 49 = 61$

$eq3 := (x-7)^2 + (y-4)^2 + z^2 = 61$

$x^2 - 14\cdot x + y^2 - 8\cdot y + z^2 + 65 = 61$

$\text{solve}\left(\begin{cases} eq1 \\ eq2 \\ eq3 \end{cases}, \{x, y, z\}\right)$

$\text{and } y = \frac{-2}{3} \text{ and } z = \frac{8}{3} \text{ or } x = 10 \text{ and } y = 8 \text{ and } z = -6$

$A(4, 5, -2)$ $B(6, 2, -3)$ and $C(7, 4, 0)$

(1) $(x-4)^2 + (y-5)^2 + (z+2)^2 = 61$

(2) $(x-6)^2 + (y-2)^2 + (z+3)^2 = 61$

(3) $(x-7)^2 + (y-4)^2 + z^2 = 61$ M1

solving (1), (2) and (3)

$x = 10, y = 8, z = -6$

or $x = \frac{4}{3}, y = -\frac{2}{3}, z = \frac{8}{3}$,

$V(10, 8, -6)$, $\left(\frac{4}{3}, -\frac{2}{3}, \frac{8}{3}\right)$ A1

both solutions are valid

g. the line through G normal to the plane is $x = \frac{17}{3} - t, \quad y = \frac{11}{3} - t, \quad z = -\frac{5}{3} + t$

$\left[\frac{17}{3} \quad \frac{11}{3} \quad \frac{-5}{3}\right] \to g$ $\left[\frac{17}{3} \quad \frac{11}{3} \quad \frac{-5}{3}\right]$

$[10 \quad 8 \quad -6] \to v1$ $[10 \quad 8 \quad -6]$

$\left[\frac{4}{3} \quad \frac{-2}{3} \quad \frac{8}{3}\right] \to v2$ $\left[\frac{4}{3} \quad \frac{-2}{3} \quad \frac{8}{3}\right]$

$\text{norm}(g - v1)$ $\frac{13\cdot\sqrt{3}}{3}$

$\text{norm}(g - v2)$ $\frac{13\cdot\sqrt{3}}{3}$

$\text{norm}(g - v1) = \text{norm}(g - v2)$ true

If $V(10, 8, -6)$, $x = 10, y = 8, z = -6$,

satisfies the line when $t = -\frac{13}{3}$

If $V\left(\frac{4}{3}, -\frac{2}{3}, \frac{8}{3}\right)$ $x = \frac{4}{3}, y = -\frac{2}{3}, z = \frac{8}{3}$,

satisfies the line when $t = \frac{13}{3}$

so both solutions for V lie on the line,

furthermore $d\left(\overrightarrow{VG}\right) = \frac{13\sqrt{3}}{3}$ A1

for both values of V

h. volume of a pyramid $V = \frac{1}{3}Ah, \quad A = \frac{7\sqrt{3}}{2}, \quad h = d(VG) = \frac{13\sqrt{3}}{3}$

$$V = \frac{1}{3} \times \frac{7\sqrt{3}}{2} \times \frac{13\sqrt{3}}{3} = \frac{91}{6}$$ A1

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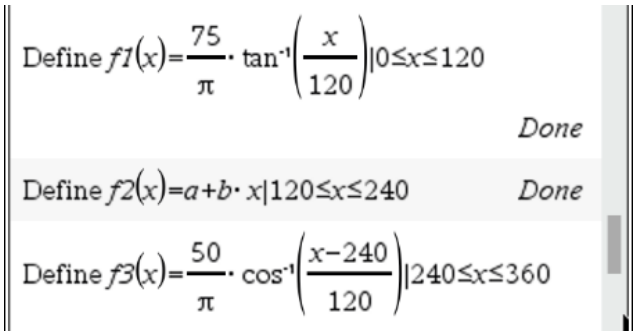
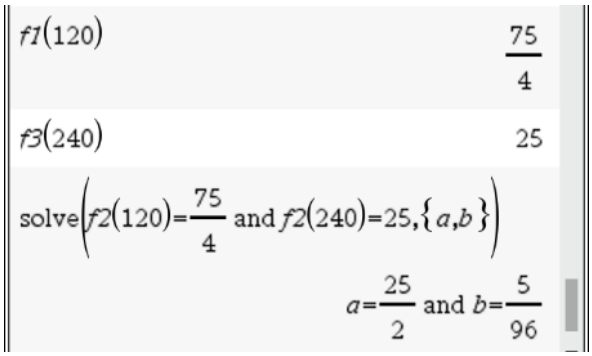
Question 4

a. $v(120)=\frac{75}{\pi}\tan^{-1}(1)=\frac{75}{\pi}\times\frac{\pi}{4}=\frac{75}{4}$ and $v(240)=\frac{50}{\pi}\cos^{-1}(0)=\frac{50}{\pi}\times\frac{\pi}{2}=25$ A1

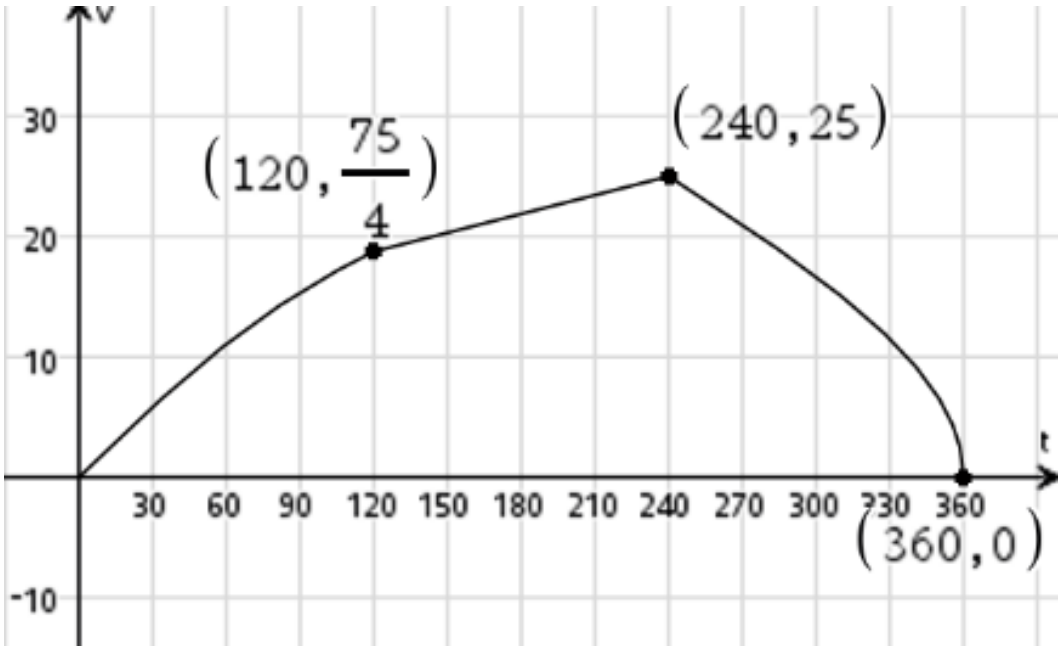
so $(1)\quad a+120b=\frac{75}{4}$ and $(2)\quad a+240b=25$ solving gives A1

$a=\frac{25}{2},\ b=\frac{5}{96}$

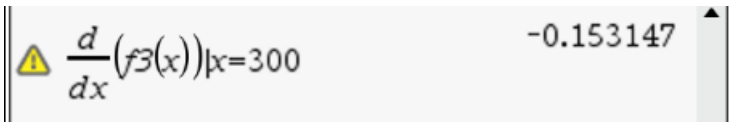




b. G2



c. when $t=300$ $\frac{d}{dt}\left(\frac{50}{\pi}\cos^{-1}\left(\frac{t-240}{120}\right)\right)=-0.153\text{ m s}^{-2}$ A1



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d. $D=\int_0^{120}\frac{75}{\pi}\tan^{-1}\left(\frac{t}{120}\right)dt+\int_{120}^{240}\left(\frac{25}{2}+\frac{5t}{96}\right)dt+\int_{240}^{360}\frac{50}{\pi}\cos^{-1}\left(\frac{t-240}{120}\right)dt$ A1

$D=5792$ m A1

e. 60 km/hr $=\frac{60\times1000}{60\times60}=\frac{50}{3}=16\frac{2}{3}$ m/s

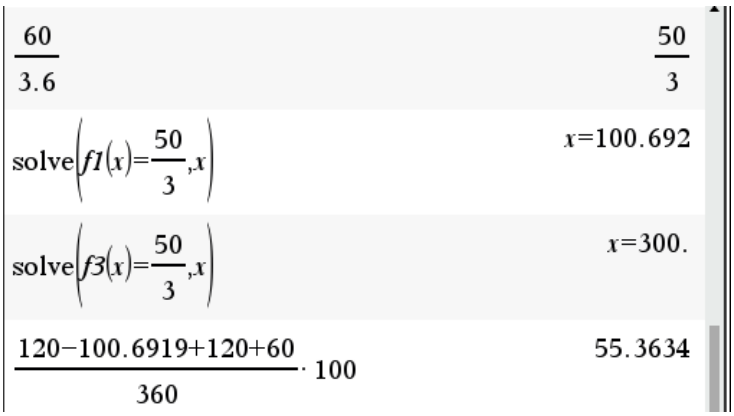
solving $\frac{75}{\pi}\tan^{-1}\left(\frac{t}{120}\right)=\frac{16}{3}$ gives $t=100.692$

solving $\frac{50}{\pi}\cos^{-1}\left(\frac{t-240}{120}\right)=\frac{16}{3}$ gives $t=300$ A1

so the percentage of the time over the speed limit is

$\left(\frac{120-100.692+120+60}{360}\right)\times100$

$=56.36\%$ A1



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Question 5

a. $\underset{\sim}{r}(t) = 18t\,\underset{\sim}{i} + 3\sin\left(\frac{\pi t}{5}\right)\underset{\sim}{j} + \left(2 + 11.7t - 5t^2\right)\underset{\sim}{k}$

hits the ground when $\underset{\sim}{r}(t).\underset{\sim}{k} = 0$ when $2 + 11.7t - 5t^2 = 0$ solving gives

$t = \frac{5}{2} = 2.5\,s$ A1

b. $\underset{\sim}{r}\left(\frac{5}{2}\right) = 45\,\underset{\sim}{i} + 3\underset{\sim}{j}$

$\left|\underset{\sim}{r}\left(\frac{5}{2}\right)\right| = \sqrt{45^2 + 3^2} = 3\sqrt{226} \approx 45.1\ \text{m s}^{-1}$ A1

Define $r(t)=\left[18\cdot t \quad 3\cdot\sin\left(\frac{\pi\cdot t}{5}\right) \quad 2+11.7\cdot t-5\cdot t^2\right]$	*Done*
Define $k=[0 \quad 0 \quad 1]$	*Done*
$\text{solve}(\text{dotP}(r(t),k)=0,t)\mid t>0$	$t=\frac{5}{2}$
$r\left(\frac{5}{2}\right)$	$[45 \quad 3 \quad 0]$
$\text{norm}\left(r\left(\frac{5}{2}\right)\right)$	45.1

c. $\dot{\underset{\sim}{r}}(t) = 18\,\underset{\sim}{i} + \frac{3\pi}{5}\cos\left(\frac{\pi t}{5}\right)\underset{\sim}{j} + (11.7 - 10t)\underset{\sim}{k}$ when the javelin hits the ground

$\dot{\underset{\sim}{r}}\left(\frac{5}{2}\right) = 18\,\underset{\sim}{i} - 13.3\underset{\sim}{k}$, $\left|\dot{\underset{\sim}{r}}\left(\frac{5}{2}\right)\right| = \sqrt{18^2 + (-13.3)^2} \approx 22.4\ \text{m s}^{-1}$ A1

the angle at which it hits the ground $\tan^{-1}\left(\frac{13.3}{18}\right) = 36.5^0$ A1

Define $v(t)=\frac{d}{dt}(r(t))$	*Done*
$v\left(\frac{5}{2}\right)$	$[18.0 \quad 0.0 \quad -13.3]$
$\text{norm}\left(v\left(\frac{5}{2}\right)\right)$	22.4
$\frac{\tan^{-1}\left(\frac{133}{180}\right)\cdot 180}{\pi}$	36.5

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d. $a_B(t) = -9.8\underset{\sim}{k}$

$\underset{\sim}{v}_B(t) = \int -9.8\,dt\,\underset{\sim}{k} = -9.8t\,\underset{\sim}{k} + \underset{\sim}{c}_1, \qquad \underset{\sim}{v}_B(0) = \frac{45}{4}\underset{\sim}{i} + \frac{3\sqrt{2}}{4}\underset{\sim}{j} = \underset{\sim}{c}_1$

$\underset{\sim}{v}_B(t) = \frac{45}{4}\underset{\sim}{i} + \frac{3\sqrt{2}}{4}\underset{\sim}{j} - 9.8t\,\underset{\sim}{k}$ M1

$\underset{\sim}{r}_B(t) = \int \frac{45}{4}dt\,\underset{\sim}{i} + \int \frac{3\sqrt{2}}{4}dt\,\underset{\sim}{j} - \int 9.8t\,dt\,\underset{\sim}{k}$

$\underset{\sim}{r}_B(t) = \frac{45t}{4}\underset{\sim}{i} + \frac{3\sqrt{2}\,t}{4}\underset{\sim}{j} - 4.9t^2\,\underset{\sim}{k} + \underset{\sim}{c}_2 \qquad \underset{\sim}{r}_B(0) = \frac{2273}{80}\underset{\sim}{k} = \underset{\sim}{c}_2$ A1

$\underset{\sim}{r}_B(t) = \frac{45t}{4}\underset{\sim}{i} + \frac{3t\sqrt{2}}{4}\underset{\sim}{j} + \left(\frac{2273}{80} - 4.9t^2\right)\underset{\sim}{k}$

e. the javelin and the ball will collide or do their paths cross?

$\underset{\sim}{i}:\ (1)\quad \frac{45t}{4} = 18s$

$\underset{\sim}{j}:\ (2)\quad \frac{3t\sqrt{2}}{4} = 3\sin\left(\frac{\pi s}{5}\right)$ M1

$\underset{\sim}{k}:\ (3)\quad \frac{2273}{80} - 4.9t^2 = 2 + 11.7s - 5s^2$

solving $(1)\,(3)$ and checks with (2) gives $t = 2$ and $s = \frac{5}{4}$ now

$\underset{\sim}{r}_B(2) = \underset{\sim}{r}\left(\frac{5}{4}\right) = \frac{45}{2}\underset{\sim}{i} + \frac{3\sqrt{2}}{2}\underset{\sim}{j} + \frac{141}{16}\underset{\sim}{k}$

so the ball and the javelin are at the same point with coordinates $\left(\frac{45}{2}, \frac{3\sqrt{2}}{2}, \frac{141}{16}\right)$

but at different times, so they do not collide but their path cross. A1

Define $b(t) = \left[\frac{45\cdot t}{4} \quad \frac{3\cdot t\cdot\sqrt{2}}{4} \quad \frac{2273}{80} - 4.9\cdot t^2\right]$ *Done*

$\text{solve}\left(\frac{45\cdot t}{4} = 18\cdot s \text{ and } 2 + 11.7\cdot s - 5\cdot s^2 = \frac{2273}{80} - 4.9\cdot t^2, \{t, s\}\right) | t > 0$

$s = \frac{5}{4}$ and $t = 2$

$b(2)$ $\left[\frac{45}{2} \quad \frac{3\cdot\sqrt{2}}{2} \quad \frac{141}{16}\right]$

$r\left(\frac{5}{4}\right)$ $\left[\frac{45}{2} \quad \frac{3\cdot\sqrt{2}}{2} \quad \frac{141}{16}\right]$

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Question 6

party pies $P \overset{d}{=} N\left(47, 3^2\right)$ sausage rolls $S \overset{d}{=} N\left(37.5, 2^2\right)$

a. total of p party pies and s sausage rolls $T = P_1 + P_2 + \ldots P_p + S_1 + S_2 + \ldots S_s = pP + sS$

$E(T) = pE(P) + sE(S) = 47p + 37.5s$

$\text{Var}(T) = p\,\text{Var}(P) + s\,\text{Var}(C) = 9p + 4s$

$\Pr(T > 1578) = 0.5 \quad \Rightarrow \quad E(T) = 1578$

$(1) \quad 47p + 37.5s = 1578$ A1

$\Pr(T < 1560) = 0.13397$

$(2) \quad -1.107819 = \dfrac{1560 - 1578}{\sqrt{9p + 4s}} = \dfrac{-18}{\sqrt{9p + 4s}}$ M1

solving (1) and (2) gives $p = 24$ and $s = 12$

two dozen party pies and one dozen sausage rolls A1

```
invNorm(0.13397,0,1)                                   -1.107819
⚠ solve(-1.107819=-18/√(9·p+4·s) and 47·p+37.5·s=1578,{p,s})
                                   s=11.999338 and p=24.000528
```

b. $H_0: \ \mu = 47$

$H_1: \ \mu < 47$ one-sided test A1

c. $n = 36, \ \bar{P} \overset{d}{=} N\left(47, \dfrac{9}{36}\right), \ \sigma_{\bar{P}} = \sqrt{\dfrac{9}{36}} = 0.5$

$p\text{ value} = \Pr\left(\bar{P} < 46.3 \mid \mu = 47\right) = 0.0808$ A1

```
normCdf(-∞,46.3,47,0.5)                                0.0808
zTest 47,3,46.3,36,-1: stat.results
                          "Title"          "z Test"
                     "Alternate Hyp"       "μ < μ0"
                            "z"            -1.4000
                          "PVal"           0.0808
                            "x̄"            46.3000
                            "n"            36.0000
                            "σ"            3.0000
```

d. Since the p value $0.0808 > 0.05$

there is no evidence to support the alternative hypothesis A1

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e. $Z \overset{d}{=} N(0,1), \quad \Pr(Z < -0.05) = -1.6449$

$$-1.6449 = \frac{p^* - 47}{0.5}$$ A1

$$p^* = 46.18$$

invNorm(0.05,0,1)	-1.6449
solve$\left(-1.6449=\frac{p-47}{0.5},p\right)$	p=46.1776

f. $n = 36, \quad \bar{S} \overset{d}{=} N\left(37.5, \frac{4}{36}\right) \quad \sigma_{\bar{S}} = \sqrt{\frac{4}{36}} = \frac{1}{3}$

$$-1.6449 = \frac{s^* - 37.5}{0.3333}$$ A1

$$s^* = 36.952$$

$$\beta = \Pr\left(\bar{S} > 36.952 \mid \mu = 36.5\right) = 0.0877$$ A1

$$8.8\%$$

solve$\left(-1.6449=\frac{sr-37.5}{0.33333},sr\right)$	sr=36.9517
sr:=36.951705483	36.9517
normCdf$\left(sr,\infty,36.5,\frac{1}{3}\right)$	0.0877

END OF SECTION B SUGGESTED ANSWERS

End of detailed answers for the
2023 Kilbaha VCE Specialist Mathematics Trial Examination 2

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